

Sharif University of Technology
Department of Computer Engineering

B.Sc. Project Report

Concise version

A Speech-to-Speech Large Language Model for Persian

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English edition of the Persian report

Abstract

This project designs, implements and evaluates a prototype Persian speech-to-speech system that operates in full-duplex mode: the user microphone stays open while the assistant speaks, and the system reacts to interruptions. Two tracks were pursued: low-rank adaptation of the end-to-end Moshika model to Persian, and a streaming cascade of NeMo speech recognition, the Qwen3-4B-Instruct-2507 responder and Piper speech synthesis. A conversational corpus of 6,754 user/response pairs (108.584 hours, session-disjoint splits) was derived from an archive of Persian spoken content.

The final cascade detects the end of speech automatically and speculatively, streams the reply clause by clause, and handles interruptions in two stages: it ducks the assistant after one frame of speech louder than the expected residual echo, then confirms lexically with the speech recognizer after discarding listener tokens and the assistant's own echoed words. In automated real-browser loopback, end-of-speech to first assistant audio had a mean of 372 ms and a 90th percentile of 431 ms, with every turn under 500 ms and the GPU side of the stack using 11.1 GiB. Interruption handling reached accuracy 0.933 and F1 0.880, with a median of 120 ms to the duck, in a deployment simulation built from real recordings, and accuracy 0.825 (F1 0.817) in the browser; an energy/F0/MFCC classifier alone reached only 0.574 on real podcast events. The responder passed all seven predeclared automatic gates of a 40-row final test (relevance 3.150 versus 1.425). All components were re-selected on fresh panels, including speech recognition on human transcripts and synthesis with two automatic listeners. A pre-registered comparison of 15 responder models and their quantized builds, judged by two models from families with no candidate, selected Gemma 4 12B at four bits for a 24 GB card (1.2 to 1.75 points of 10 above the best 4B model, 39 ms to the first token); in the live stack it met the memory and latency limits but fell short of the coherence rule after late interruptions, so it remains an opt-in preset. For a 12 GB card, post-hoc checks of the same model (including a llama.cpp setting that cut its first token from about 180 to 25 ms) failed the same coherence rule.

The direct track produced no deployable model. Moshi's English tokenizer needs 6.6 tokens per Persian word and exceeds its one-token-per-frame budget; phoneme romanization doubled the best runtime result but the speech was still not intelligible, and a speech-input Gemma 4 model also underperformed the cascade. Latency and interruption evidence is automated and on an H100 (the memory of a 12 GB card was respected, its compute speed was not), labels are automatic, and the results of the human listening study are not yet available.

Keywords: Speech-to-speech, full-duplex dialogue, barge-in detection, Persian language, large language models, streaming cascade, tokenization, parameter-efficient adaptation

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Chapter 1

Introduction

This document is the **concise edition** of the B.Sc. project report. The full report, with seven chapters, a glossary and a reproduction appendix, is a separate file. Numbers, requirements and claim boundaries are the same in both editions; what is shortened is explanation and repetition, not results.

Spoken conversation is the most natural form of human–machine interaction, but until recently it was almost always built from a chain of independent components. Models such as Moshi [1] and Freeze-Omni [2] showed that dialogue can be modelled as a single "speech-to-speech" problem, with one model that listens and speaks at the same time. This shift has happened for high-resource languages. For Persian, none of the open models offers official support, and no conversational corpus with interruption labels is publicly available.

The practical question of the project is: with the budget of a bachelor's project, how far can one go in building a full-duplex Persian speech-to-speech system, and what can be "proven" with auditable evidence?

1.1 Problem Definition

The defined goal is to design, implement and evaluate a prototype speech-to-speech large language model for Persian. The system must be full duplex: the user's microphone stays active while the reply plays, and the system must be able to react to the user's "interruption" (barge-in). Three properties separate this problem from conventional spoken question answering: listening and speaking at the same time; a decision to stop playback within a few tens of milliseconds; and continuity with what the user says at the moment of the interruption, so that the user does not have to repeat the sentence.

1.2 Motivation

A speech interface can be the only effective way to use digital services for users who are not comfortable with a Persian keyboard, for people with low vision, and in particular for older adults. Persian speech recognition and text-to-speech resources exist; a prominent example is ManaTTS [3], with more than 86 hours of single-speaker speech. These resources are read speech, however, and do not cover overlap, interruptions or backchannels. On the other hand, current models such as Qwen2.5-Omni [4] need far more memory for their 7-billion parameter version than the 12–24 GB range. Showing what is and is not achievable within this range is itself an engineering result.

1.3 Defined Requirements

The project definition sets seven specific requirements (Table 1.1). The expected outputs are four items: a working prototype, documented source code, a Persian conversational dataset, and findings on interaction with older adults. For the fourth item, the measurement protocol and infrastructure are ready; its empirical finding can only be claimed once the human study is complete, so this report makes no claim about the experience of older adults.

Table 1.1: The seven requirements of the project definition

ID	Requirement	Specified measure
R1	Persian speech-to-speech prototype	Persian spoken reply generated from Persian spoken input
R2	Full-duplex operation	Listening during playback and reacting to interruptions
R3	End-to-end latency	At most 500 ms from receiving speech to producing the spoken reply
R4	Persian conversational dataset	100–200 hours of speech with noise, overlap and interruptions
R5	Interruption detection module	Accuracy above 80% with energy, pitch and MFCC features
R6	Evaluation on target hardware	A GPU with 12–24 GB of memory
R7	Human evaluation	5–10 Persian speakers, at least two aged over 60

1.4 Two-Track Strategy and Challenges

Four challenges determined the final shape of the work: the lack of an official training path for Persian in models such as Mini-Omni2 [5] and LLaMA-Omni2 [6]; the lack of a labelled

Persian conversational corpus; the gap between reducing offline loss and valid autoregressive generation; and the boundary between engineering evidence and scientific claims.

To reduce risk, two tracks were pursued in parallel. The direct track is low-rank adaptation of Moshika with the official trainer of the Moshi family [7]. The cascade track is a chain of Persian speech recognition, a text language model and text-to-speech, on top of which full-duplex control and interruption logic are built separately. Both tracks used the same dataset. The outcome was asymmetric: the cascade track reached a streaming full-duplex system that meets the numeric latency and interruption criteria in automated tests, and the direct track is reported as a negative result together with its measured cause.

1.5 Contributions

1. **A streaming full-duplex Persian voice assistant.** Open microphone, speculative end-of-speech detection and clause-by-clause playback; in browser loopback, a mean latency of 372 ms and a 90th percentile of 431 ms with 11.1 GiB of GPU memory.

2. **Two-stage interruption detection.** Ducking after one frame and a lexical confirmation with echo rejection; accuracy 0.933/F1 0.880 in simulation and 0.825 in the browser, against 0.574 for an acoustic classifier alone.

3. **An audited dataset.** 6,754 pairs with 108.584 hours of audio, session-based splits and zero group leakage.

4. **Evidence-based component selection.** A final semantic test (3.150 versus 1.425) and re-selection of the recognizer, responder and synthesizer on fresh panels, including a comparison of 15 responder models and their quantized builds for 12 GiB and 24 GB cards.

5. **A measured cause for the failure of the direct track.** Seven generations of Moshika and one speech-input model; the tokenizer bottleneck is real but not the only obstacle.

6. **An evidence-first method.** Protocols were registered before running, test splits were opened once, and artifacts are write-once.

Latency and interruption evidence is automated and on an H100, and the results of the listening study are not yet available.

1.6 Structure of This Report

Chapter 2 condenses the necessary concepts and the position of the work. Chapter 3 describes how the data were built, the cascade architecture, the interruption detector, the direct-track

experiments and the evidence classes. Chapter 4 reports the measured results with the claim boundary of each. Chapter 5 summarizes the interpretation, limitations and future work.

Chapter 2

Background and Key Concepts

This chapter gives only the concepts needed to read the method and the results, and positions the project against prior work.

2.1 Two Families of Architectures

Spoken dialogue systems fall into two families. In the **cascade** family, speech recognition, a text dialogue engine and text-to-speech run separately. This architecture is auditable and replaceable, but component latencies add up and each stage's errors propagate to the next. In the **end-to-end** family, the model goes directly from speech to speech. Natively full-duplex models such as Moshi model two audio streams in parallel and need no explicit speech segmentation for turn-taking.

The project definition is end-to-end in spirit, but the bachelor's budget and the lack of an official Persian trainer made the cascade unavoidable as the delivered system.

2.2 Language Models, Low-Rank Adaptation and Exposure Bias

The project's text responder is an instruction-following model. Full fine-tuning of a multi-billion-parameter model on limited hardware was not practical, so the direct track used LoRA [8]: the weight update is the product of two low-rank matrices added to the frozen weight.

The decisive point for interpreting the direct-track results is the difference between training and inference. In training, the model sees the previous reference token at every step; this is **teacher forcing**, and the reported loss is computed under it. At inference, the model must condition on its own generated tokens. If the two distributions drift apart, errors accumulate. This mismatch is called **exposure bias** [9], and its standard remedy is scheduled sampling [10]. The

practical consequence is explicit: lower teacher-forced loss does not guarantee valid autoregressive generation.

2.3 Recognition, Synthesis and Interruption Features

The system's speech recognizer is a hybrid RNNT-CTC model of the Conformer family, run locally with the NeMo stack [11]. Text-to-speech uses the Piper engine [12] with a Persian voice.

The interruption detector computes classical features on a rolling window: energy, zero-crossing rate, MFCCs [13] and the fundamental frequency by YIN [14]. The classifier is gradient boosting [15]. Confidence intervals for accuracy are reported with the Wilson method [16] and block-wise at the session level. In human conversation, turn-taking gaps are often around 200 ms [17], so the interruption decision must be made much faster than a complete reply is generated.

2.4 Position Among Prior Work

Mini-Omni2 and LLaMA-Omni2 have inference repositories but provide no complete, official speech-to-speech trainer. Freeze-Omni is full duplex, but its trainer is incomplete for a new language. Qwen2.5-Omni does not list Persian among its spoken input and output languages, and its 7-billion-parameter version is outside the target memory range. PersonaPlex [18], built on Moshi, conditions role and voice, but at the time of this project's decision no official, available trainer for a new language existed. Among the models reviewed, only the Moshi family met all three conditions of the direct track: native full duplex, open weights, and an official low-rank adaptation trainer.

For Persian, single-speaker read-speech resources are no substitute for an interactive conversational corpus. Table 2.1 summarizes the comparison.

2.5 Measures and Model Judging

The mechanical quality of the chain is measured by whether a row passes the conditions of a Persian transcript, a Persian reply and non-silent audio. Content preservation of the generated speech is approximated by re-recognition and character and word error rates; this measure is not human intelligibility. The semantic quality of replies was measured by an LLM judge on relevance and coherence, double-blind and with two repetitions. Because the judge and the candidate are of the same family, the result stays in the class of "same-family automatic approximation" and is no substitute for human evaluation or the ITU-T P.800 standard [19]. Training,

Table 2.1: This project compared with prior work on the capabilities of the project definition

Work	Full duplex	Open weights	Official trainer	Persian support	Memory range
Mini-Omni2	command-based	yes	no	no	fits
LLaMA-Omni2	no	yes	no	no	fits
Freeze-Omni	yes	yes	partial	no	fits
Qwen2.5-Omni	yes	yes	no	no	does not fit
Moshi	native	yes	yes	no	does not fit
PersonaPlex	native	yes	partial	no	does not fit
This project, cascade	external control	yes	not needed	yes	fits
This project, direct	native	yes	yes	failed	does not fit

The "memory range" column is the fit to the 12–24 GB of the project definition, not the scientific quality of a model. "Failed" for the direct track means failing the autoregressive runtime condition, not a loss that did not decrease.

validation and test splits are always based on the source session identifier, so that no group leakage occurs.

Chapter 3

Proposed Method and Implementation

This chapter describes the structure of the delivered system, the data pipeline, the interruption detector, the design of the direct-track experiments, and the evidence-first way of working.

3.1 Early Attempt and Revision

The first implementation was based on LLaMA-Omni2: a frozen Whisper encoder, a projector, the Qwen2.5-0.5B [20] language model with a LoRA adapter, and separate recognition and generation heads. The loss decreased, but an internal audit found three fundamental defects: the language model did not take part in the forward pass; the recognition target used per-process random hashes; and the generation target was reconstruction of the input utterance, not a conversational reply. That artifact was not a speech-to-speech model. Two decisions followed from this audit: serving moved to a cascade with real components, and for the direct track only a model with an official trainer was accepted.

3.2 Dataset Construction

No public Persian corpus with interruption labels was available. The dataset was derived from an internal archive of spoken content from four channels. Internal research use was recorded with the supervisor's approval; this approval is not a licence to redistribute the raw audio or captions, and no restricted data are in the code repository.

In the first design, text labels were produced by re-running the speech recognition service. That same service had earlier been tuned on the captions of these channels, which created a circular loop. Once this was identified, the text labels were switched to the original source captions, and recognition was kept only as a runtime component and a measurement tool.

The pipeline is: a caption inventory of 775.887 hours, weighted programme selection, reconstruction of fifteen-minute windows, speaker diarization, noise condition labels, extraction

of "user speech / response" pairs without reuse, and an audited stereo export with channel zero for the reconstructed assistant and channel one for the user. Train/validation/test splits are by source session identifier. The interaction labels are automatic candidates, and the number of human-verified rows is zero.

3.3 Architecture of the Streaming Cascade

The delivered system is a streaming cascade with full-duplex control (Figure 3.1). The first version worked with a push-to-talk button and built the whole reply before playback (about 2.2 seconds to the first audio). In the final version the stages overlap: the end of speech is detected automatically, generation starts speculatively, tokens arrive as a stream, and each clause is synthesized and played as soon as it is complete.

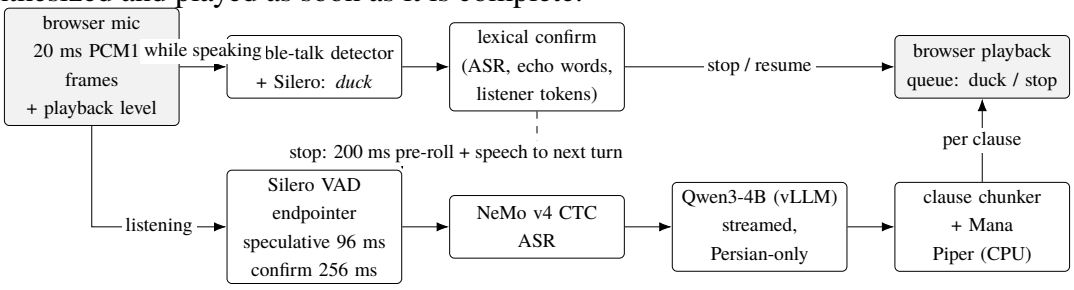


Figure 3.1: The streaming architecture. The bottom row is reply generation while listening; the top row is interruption control while the assistant speaks.

The browser client captures the microphone continuously and sends one frame every 20 ms (a sequence number, the level of the assistant audio being played, and 320 PCM16 samples); it schedules the reply clauses on a playback queue and reports timings with its own clock. The Silero detector starts a speculative turn after 96 ms of silence and holds its audio until the end of speech is confirmed at 256 ms; if the user continues, the turn is cancelled and the speech is added to the same turn. Recognition uses NeMo version 4 (CTC), which the candidate comparison found to be the most accurate candidate within the latency budget. The reply is generated by Qwen3-4B-Instruct-2507 [21] with the locked v2 prompt, streamed from vLLM with two turns of history, and a "Persian script only" constraint is applied during decoding through structured output. The text is cut at clause boundaries (a first fragment of 2–4 words, then at most 14 words), and each fragment is synthesized with Piper and the Mana voice on the CPU.

The nine-row mechanical panel and the earlier re-recognition measure were run on the chain before the upgrade (Qwen2.5-0.5B-Instruct); the 40-row semantic test is on Qwen3-4B with the Qwen3.8-27B-FP8 judge.

3.4 Interruption Detection and the Full-Duplex Layer

While the assistant speaks, a two-stage controller inspects the frames. Stage one: a frame must be speech and louder, by a margin, than the expected echo of the assistant's voice (the playback level reported by the client plus an estimated coupling factor, assumed conservatively at the start of each reply and then adapted); one such frame ducks the assistant, giving a "detect and adapt" reaction with a 150 ms target. Stage two: the speech captured since the onset of the interruption is transcribed by the same recognizer; listener tokens (آره, بله, اوهوم, …), words of the reply being played (residual echo) and function words are discarded. If any word remains, playback stops, generation is cancelled, and the user's speech, with a 200 ms pre-roll, becomes the next turn; otherwise the volume returns and the reply continues. Development runs of the simulation showed that a confirmation based on speech duration or on an acoustic classifier does not separate long backchannels, loud noise and echo from interruptions.

Evaluation has three levels: a classifier test on podcast events with independent two-pass automatic Gemini labels, a deployment simulation built from real recordings, and loopback testing with a real browser. The test backchannels were selected with a different recognition model (SeamlessM4T v2), so that the lexical confirmation is not circular.

3.5 Design of the Direct-Track Experiments

The direct track is low-rank adaptation of Moshika with the official trainer. The input is the stereo export of Section 3.2. The loss is a weighted combination of the text loss and the audio-token loss.

The runtime acceptance condition was locked before training: the adapter is loaded in the official streaming server, and for each of nine predefined rows there must be text, non-silent speech and the Persian-script constraint. A checkpoint is eligible only with 9/9. The condition is deliberately strict, because the lowest validation loss in the first version gave almost silent audio without text.

The experiments are a chain of falsifiable hypotheses, and in each generation mostly one factor changed (Table 3.1). Version v6.2 is a capacity test, not a generalization test: training and evaluation on the same 32 samples, a 12-second window, and scheduled dropout of the text input from 0.25 to 0.75. If the loss does not decrease even on training data, the problem is capacity; if it decreases but runtime fails, the problem is exposure bias.

The final audit revealed a structural cause that the first six generations had not tested: Moshi's English tokenizer needs 6.6 tokens per Persian word (55% byte fallback), while the model produces at most one text token per 12.5 Hz frame; 98.5% of turns exceeded this budget. Version v7a made the inner-monologue text a romanization of the phonemes Piper actually spoke (2.7 tokens per word) with word-by-word timing, and kept the other conditions of the

Table 3.1: Design of the seven experimental generations of the direct track

Version	Changed factor	Tested hypothesis	Rank
v1	base configuration, 20-second window	the model learns Persian speech with enough data	64
v2	100-second text window	a short window does not fit the start of the reply in context	64
v3	frozen embeddings, higher rank	training the embeddings damages the base representation	128
v4	only the two text embeddings	only the Persian text space needs adaptation	128
v5	first-codebook loss weight from 100 to 10	a heavy semantic-codebook weight stifles acoustic learning	128
v6.2	in-sample capacity test, 12-second window	the model has capacity and the problem is exposure bias	64
v7a	romanized phoneme text, word-level timing	the tokenizer breaks the one-token-per-frame budget	64

capacity test fixed. The second direct track is Gemma 4 E4B, which hears audio directly; it was measured against the cascade zero-shot and with a DiVA-style self-distillation LoRA adapter (3,817 turns, 11.2 hours).

3.6 Evidence-First Method

Five operating rules were applied. One: the protocol, row indices and thresholds are registered before running. Two: semantic evaluation has two stages; only after the seven-row qualification passes completely is the final test opened, once. The Qwen3.8-27B-FP8 judge runs at temperature zero, in blind order and with two repetitions. Three: result artifacts are write-once. Four: each measurement has an evidence class (Table 3.2), and only the official class can prove a requirement of the project definition. Five: the aggregator fails closed; a missing artifact or a class below what is needed makes a condition fail, not "unknown".

Table 3.2: Evidence classes and the claim power of each class

Evidence class	Required conditions	Proves a requirement
Official end-to-end	live consented session, client playback confirmation, real microphone and eligible hardware	yes
Unofficial live	same protocol, without eligible hardware or study conditions	no
Server-side component	generation or detector timing, without proof of the client's audible behaviour	no
Synthetic approximation	generated audio or simulated events	regression test only
Same-family automatic approximation	judging by an LLM of the candidate's family	no

The consequence of this method is that the evidence status of each condition is stated independently of whether the report is ready for presentation and examination. The final result is a clear separation of what was built from what was proven with official evidence.

Chapter 4

Experiments and Results

For each result, the value, the denominator and the permitted claim boundary are given. All numbers are read from recorded artifacts.

4.1 Coverage of Acceptance Conditions and Evidence Status

Table 4.1 places the official acceptance conditions (human labels, target hardware, a completed study) next to the automatic evidence of the final stage; the automatic evidence does not replace the official conditions.

Table 4.1: Official acceptance conditions and the final automatic evidence

Acceptance condition	Official	Final automatic evidence
Working Persian speech-to-speech prototype	sufficient	streaming system and loopback test
Final automatic semantic test	sufficient	–
Audited 100–200 hour export	sufficient	–
Data policy under a documented waiver	sufficient	–
Source code licence	sufficient	–
Deployable direct model	negative	v7a: 6/9, unintelligible
Detector above 80%	not human	simulation 0.933/0.880; browser 0.825/0.817
Latency and memory on target hardware	not target hardware	11.1 GiB; mean 372, 90th percentile 431
Human study	pending	registered instrument

4.2 Dataset

Table 4.2 shows the final characteristics. The 108.584 hours fall within the range of R4, and the automatic integrity audit, including channel order, hashing and group splits, passed without failure.

Table 4.2: Characteristics of the derived Persian conversational dataset

Indicator	Value
Caption inventory of the channels	775.887 hours
Response pairs without reuse	6,754
Source audio hours of the pairs	123.796 hours
Speakers / sessions	407/162
Exported stereo hours	108.584 hours
Train / validation / test split	6,419/131/204
Group leakage / automatic audit failures	0/0
Automatic interaction-event candidates	770
Human-verified rows	0

An independent automatic listener (Gemini, two passes, agreed labels only) on 300 available pairs heard another speaker start at 93.8% of turn boundaries and confirmed 75.7% of the marked overlaps; the captions differ from its transcript by a 15.9% character error rate.

The interaction labels are automatic candidates and their verification is automatic too. This dataset cannot be called human-labelled, and the source data are not redistributable.

4.3 Cascade System: Mechanical Panel and Intelligibility

The validation panel is nine evenly spaced rows of the 131-row manifest. The row indices were locked before running. The chain is the real recognition service, the Qwen2.5-0.5B-Instruct responder and the real speech engine, with the audited user channel one as input. This panel measures the mechanics of the chain, not the semantic quality of Qwen3-4B. The result is in Table 4.3.

The median, 95th percentile and maximum transcript lengths were 445, 3977.8 and 5031 characters. The descriptive Pearson correlation between transcript length and full generation time is 0.885; the longest transcript and the slowest turn were both row 32. The time column is the duration of full reply generation, not first-audio latency.

Re-recognition of the same nine outputs, with the reply text as reference, gave a micro-aggregated character error rate of 7.14% (47/658) and a word error rate of 30.41% (52/171). No row had an exact word or character match. Figure 4.1 shows both axes.

The mechanical result is out-of-sample with the Qwen2.5-0.5B responder and is not generalized to Qwen3-4B. Nine rows are not enough for population generalization. Full generation time is not the T_{first_audio} measure and does not meet the 500 ms condition. Re-recognition does not prove human intelligibility or naturalness.

Table 4.3: Result of the fixed validation panel of the cascade

Indicator	Value
Text responder of the panel	Qwen2.5-0.5B
Rows passing every condition	9/9
Rows with a rule-based fallback reply	0/9
Minimum Persian-character share in transcript and reply	1.000
Non-silent audio replies	9/9
Minimum RMS of the reply audio	0.133529
Median / maximum full generation time (ms)	5584.7/13270.2
Final-test rows used	0

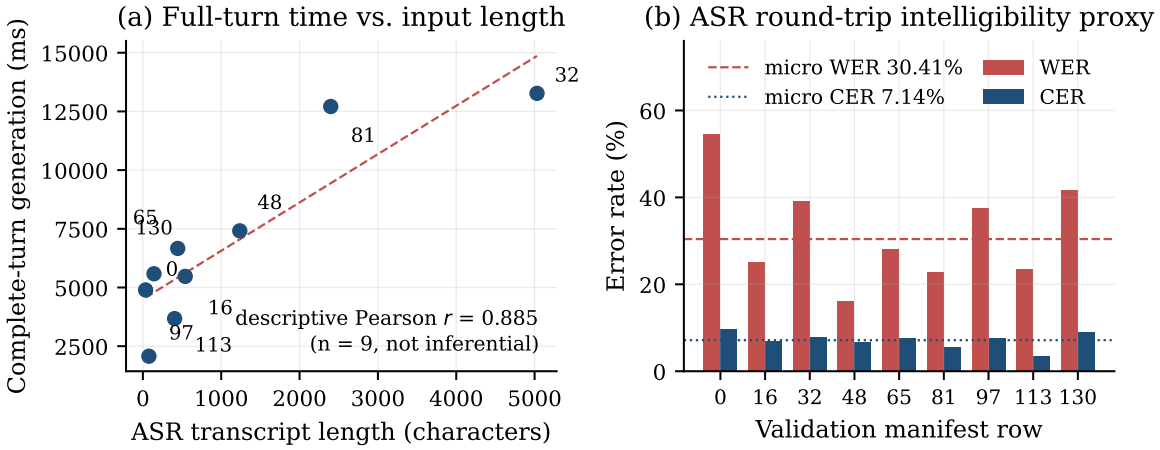


Figure 4.1: (a) Full generation time against transcript length for the nine panel rows. (b) Word and character error rates of re-recognizing the same nine outputs.

4.4 Automatic Semantic Test

The qualification stage on seven unused validation rows passed every condition: 42/42 valid calls, relevance 3.857 versus 1.000, coherence 3.571, paired wins 6/7, and relevance of at least two in 100% of rows. Only after this pass was the final test opened, once.

The final test ran on 40 rows of the 204-row manifest from five held-out sessions. Three blind arms: the locked Qwen2.5-0.5B baseline, the Qwen3-4B candidate with prompt v2, and the automatically aligned next-turn reference. The judge was Qwen3.8-27B-FP8 at temperature zero with two repetitions per row. Table 4.4 shows the result and Table 4.5 the passing of all seven predefined conditions.

Figure 4.2 shows the means and the distribution of scores. The low score of the aligned reference shows that "saying the corpus's next turn" is not necessarily an appropriate reply to the user's utterance.

Table 4.4: Result of the final automatic semantic test on 40 group-disjoint rows

Indicator	Value
Valid / expected judge calls	240/240
Judge repetition agreement	1.00
Mean relevance candidate / baseline / reference	3.150/1.425/0.950
Mean coherence candidate / baseline	3.325/1.525
Relevance / coherence gain over the baseline	+1.725/+1.800
Paired relevance wins	27/40 (67.5%)
Relevance of at least two	35/40 (87.5%)
Candidate generation retries / invalid audio	0/0

Table 4.5: Predefined automatic conditions of the final test

Condition	Threshold	Observed	Status
Candidate mean relevance	≥ 2.00	3.150	met
Candidate mean coherence	≥ 2.50	3.325	met
Relevance gain	≥ 0.50	1.725	met
Coherence gain	≥ 0	1.800	met
Paired relevance win rate	≥ 0.60	0.675	met
Rate of relevance at least two	≥ 0.70	0.875	met
Full mechanical validity	240/240	240/240	met

A side LoRA experiment on the responder reduced the development loss by 31.18% but did not beat the baseline in the semantic evaluation and was not used in the final system.
Passing seven automatic conditions does not mean perceptual quality or an independent benchmark. The judge and the candidate are of the same family.

4.5 Component Re-Selection

All three components were compared again under new protocols. **Recognition:** on 344 PSRB clips with human transcripts, Seamless was the most accurate (character error rate 6.92%) but, with a median of 671 ms, exceeded the 150 ms budget; NeMo version 4, with 9.44% and 58 ms, was the best candidate within the budget and was retained (Whisper large-v3, Qwen3-ASR and Gemma 4 above 20%). A second round added eight Persian checkpoints: Shenava Koochik was 0.50 points better, not significantly, and Rade-ASR-CTC-3B reached 16.1%, mostly from speech it left out. **Responder:** on a panel of 60 elderly requests and 60 natural turns, no candidate reached the replacement threshold (+0.25 relevance) under both judges. The judges disagree: under Gemini, Qwen3.5-4B and Gemma 4 with the elderly prompt are about one point better

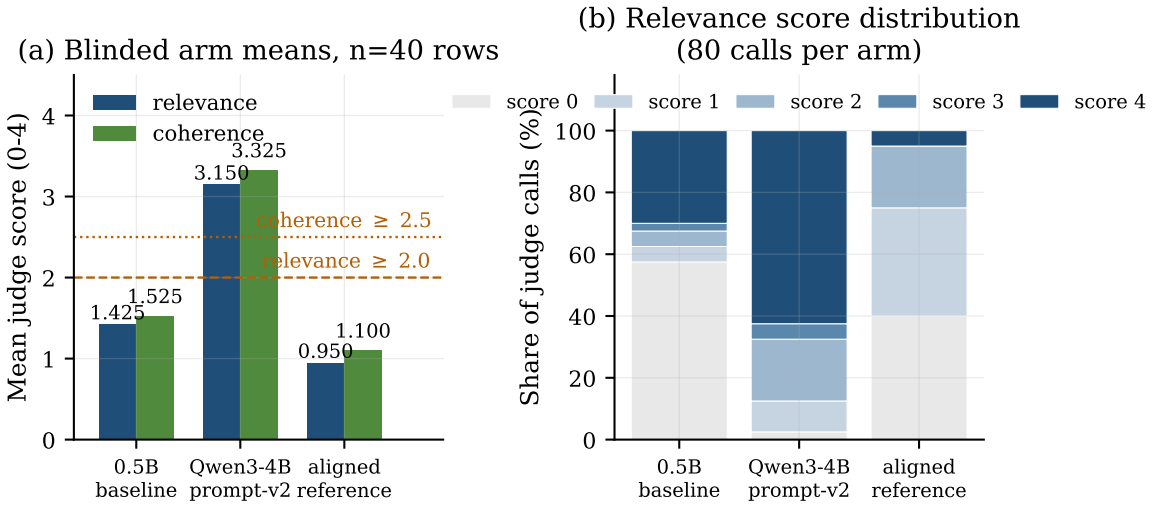


Figure 4.2: (a) Mean relevance and coherence of the three blind arms. (b) Distribution of the relevance score over the 80 judge calls of each arm.

than the deployed system (3.36 and 3.54 versus 2.38), but not under the Qwen judge (3.31 and 3.00 versus 3.29). The Qwen3.5-4B configuration passed the pre-registered deployment qualification (memory 11.1 GiB, mean and 90th-percentile latency 345 and 398 ms, coherence after interruption under both judges); its replacement depends on the preference of the listening-study participants. A pre-registered size comparison then served five larger models of 9 to 36 billion parameters, most at four bits, under a 22 GiB limit. Gemma 4 31B did not fit, and none of the other four beat Qwen3.5-4B under both judges. Under the Qwen judge none gained more than 0.01 in relevance. Under Gemini, Gemma 4 12B gained 0.26 but won only 26% of the rows. On this panel, size did not buy reply quality. A third pre-registered comparison then covered 15 models of the Qwen3.5, Qwen3.6, Qwen3.8 and Gemma 4 families (0.8 to 35 billion parameters) with every quantized build that vLLM and llama.cpp can run, for a 12 GiB and a 24 GB card (responder within 10,682 and 22,528 MiB, a 2,048-token context and two concurrent requests, a median first token of at most 150 ms), scored from 0 to 10 on 252 new test rows by two judges from families with no candidate, Claude Opus 5.5 and GPT-6 Astra. For 24 GB, Gemma 4 12B with Google's 4-bit weights on vLLM won (6.47 and 7.09; first token 39 ms; 10.9 GB) and beat Qwen3.5-4B (5.27 and 5.34) by 1.20 and 1.75 points and the deployed responder (2.84 and 3.58) by 3.63 and 3.52, all with 95% intervals above zero. In the live stack it met the memory (12.2 GiB) and latency limits (mean 372, 90th percentile 468 ms) but not the coherence rule after interruption (0.37 points below the controls under both judges, limit 0.25): when the recognizer garbled a follow-up it asked the user to repeat (14 of 60 replies, against 3 of 60 for Qwen3.5-4B). It is not qualified. For 12 GiB no configuration both fit and met the latency gate. Two post-hoc 12 GiB checks of the same model failed as well: vLLM with one sequence (one unanswered utterance, coherence under the Qwen judge) and llama.cpp after a diagnosis traced its slow first token to cache checkpoints of the sliding window (24.8 ms without them; it then failed latency under other users' load and coherence under both judges). The two

judges agree with each other (rank correlation 0.88); more judge reasoning and repeated latency measurements with the GPU load recorded changed no result. A first decision, the FP8 build of the same model with equal test quality, was revised after two builds the rules required were found unrun; both are reported. With judges from outside the candidate families, size did buy reply quality. **Voice:** of the six Persian Piper voices, Mana remained the most intelligible to two automatic listeners, with a mean character error rate of 5.82%; ganji_adabi was tied and had the best DNSMOS, and both went to the listening study. ParsVoice-XTTS with the ManaTTS voice was on par on a typical clip but broke about one clip in ten (mean error 13.55%, significantly worse), and its first streaming chunk took 307 ms; it did not enter the study. The lab VITS model, even after inference optimization without retraining (error from 9.89% to 9.19%, the highest DNSMOS), stayed significantly worse than Mana and did not enter the study.

4.6 Interruption Detection

The deployed controller was measured in a deployment simulation: the microphone during playback, residual echo of the reply, background noise, a room response and one real event (an interruption, a backchannel, noise or nothing). Six development runs used only the training split: a duration-and-classifier confirmation 0.687, the lexical confirmation 0.840, and with corrected initialization of the double-talk detector and echo adaptation 0.920. The chosen variant reached, once, on 600 test scenarios an accuracy of **0.933** and an F1 of **0.880** (recall 0.98; wrong stops on backchannels 8.7%, noise 9.3% and no event 6.7%), with a median of 120 ms from the acoustic onset to the duck. The energy/pitch/MFCC classifier reached only an accuracy of 0.574 and an F1 of 0.672 on 94 podcast test events with Gemini labels.

A simulated mixture built from real recordings, with automatic labels; not a live user.

4.7 Direct Track

Table 4.6 summarizes the seven generations. The common pattern: the loss decreases and the runtime condition is not met. The minimum losses are not directly comparable across generations, because the window and the evaluation set differ. For v6.2 and v7a the loss is in-sample.

In v6.2 the best text loss after step 50 is 0.550852, 32.20% lower than the 0.812474 at step 50 and above the 30% criterion; that is, the adapter has the capacity to fit the target. Nevertheless all four checkpoints scored 0 of 9, and in each checkpoint only one of the nine rows produced text and speech together. Figure 4.3 sets these two axes against each other.

No checkpoint is eligible for deployment. The statement "Moshika was successfully fine-tuned for Persian" is false on this evidence.

Version seven. With romanized text and word-by-word timing, the best accepted count in the same capacity test reached 6/9 (controlled baseline 3/9), and the server produced audible

Table 4.6: Summary of the seven generations of the direct track. "Best run" is the largest number of the nine rows passed among the checkpoints of that generation.

Version	Window (s)	Candidates	Minimum loss	Best run	Eligible
v1	20	16	1.402	0/9	no
v2	100	20	1.735	2/9	no
v3	100	5	1.879	1/9	no
v4	100	5	1.879	1/9	no
v5	100	5	1.868	1/9	no
v6.2	12	4	2.060	0/9	no
v7a	12	4	1.264	6/9	no

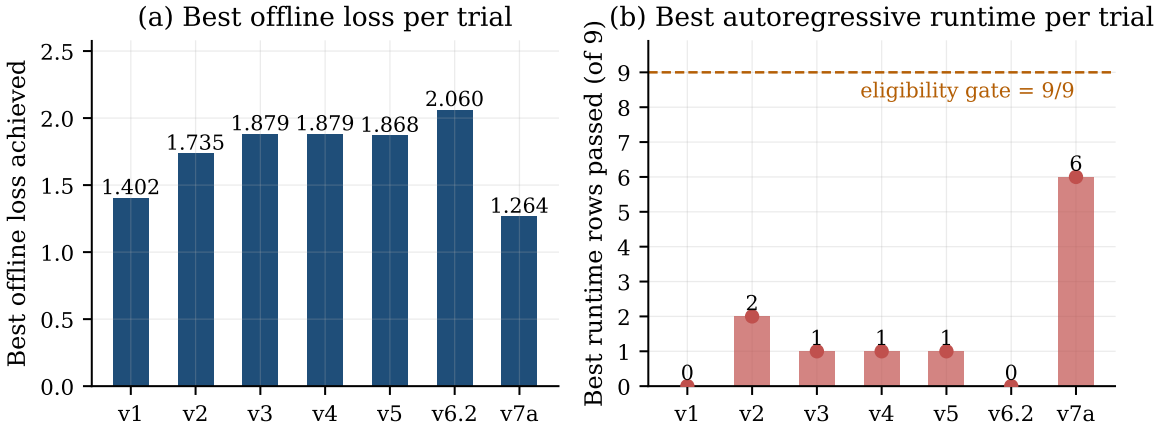


Figure 4.3: (a) Best offline loss of each generation. (b) Largest number of rows passing the runtime panel, against the 9/9 condition.

speech with text on 8 of 9 rows; but the 9/9 condition was not met and the generated speech was not intelligible to a Persian recognizer (mean prefix error 0.88), so full training was not started. The tokenizer bottleneck is real but not the only obstacle.

Speech input. Gemma 4, hearing the audio directly, scored 1.98 and 2.05 in relevance zero-shot and with the adapter, against 2.80 for the cascade; the result is negative by the pre-registered rule. Removing recognition, even with the same language model, lowered quality (2.49 versus 1.98) and saved only about 20 ms. With the second judge (Gemini) the deployed system scored lowest (1.91) and the speech-input path higher (2.18, and 2.52 with the adapter); the official result with the first judge stays negative, and this disagreement between the judges is reported. Gemma 4 12B understood Persian speech even worse than E4B.

4.8 Latency, Memory and Browser Loopback

A headless browser ran the real client page with a fake microphone (recorded utterances of 1–5 seconds). The language model server was limited to the memory share of a 12 GB card: the peak GPU memory of the whole system was 11.1 GiB (at H100 speed). With the speculative start after 96 ms of silence (chosen on development utterances), no turn of the 201 turns of the main panel went unanswered; the mean latency from end of speech to first audio was **372**, the 90th percentile **431** and the maximum 491 ms (the first run with 160 ms: 418, 487 and 537); with the browser's output latency, the 90th percentile is 473. Simulated elderly speech: mean 382 and 90th percentile 446. In 120 turns with interruptions and backchannels, the decision accuracy was 0.825, the F1 0.817, and the median time to duck from the detected onset 44 ms. In 30 scripted dialogues with a follow-up question over the reply, 24 of 27 interruptions stopped the reply, and the reply after an interruption was as relevant as the reply to a normal turn under both judges. A semantic end-of-turn gate (Smart Turn v3.2 [22], an LLM judgement, a lexical rule and a Smart Turn fine-tuned on Persian) lowered premature endpointing on development utterances from 19.8%, but no candidate delayed fewer than 13% of genuine turn ends (pre-registered limit 5%). A second round labelled each development clip's ending (Gemini, two passes) and counted only complete endings: the LLM judgement then qualified (13.3% premature, 4.4% of complete endings held), but in the loopback test it held 15% of finished turns and the 90th percentile rose to 656 ms, so it was removed; the deployed endpointer remained unchanged.

Automated loopback on an H100 without a human user or a physical audio path; the proposal's criterion (mean and 90th percentile) and, on the main panel, the every-turn reading are met.

The small listening study (naturalness of two Piper voices and human speech, three recorded dialogues, and a preference between the replies of two responders that, by a pre-registered rule, decides on replacing the responder) is ready with a registered protocol; its results are not yet available.

4.9 Final Status of the Requirements

The evidence for R2, R3 and R5 is automated and on an H100; R4 has no human labels; R5 was not met by the specified acoustic features alone.

Of the four expected outputs, three have been produced and documented: the streaming cascade prototype, documented source code, and the Persian conversational dataset. Empirical findings on older adults can be reported once the listening-study responses arrive.

Table 4.7: Final status of the seven requirements of the project definition

ID	Status	Supporting evidence
R1	met (streaming cascade)	final semantic test; all 401 loopback turns answered
R2	met in automated tests	interruptions 0.83/0.82 in the browser, 0.93/0.88 in simulation
R3	met	mean 372 and 90th percentile 431; maximum 491 ms
R4	met	108.584 hours of audited export (automatic labels)
R5	goal met by another method	lexical confirmation; acoustic classifier alone 0.574
R6	declared deviation	memory 11.1 GiB; speed on an H100
R7	awaiting results	registered instrument and protocol

Chapter 5

Discussion, Conclusions and Future Work

5.1 The Gap Between Loss and Generation

The main scientific finding of the direct track is that reducing loss under teacher forcing is separate from valid autoregressive generation. Version v6.2 sharpens this separation: on the same 32 training samples, the text loss fell by 32.20%, but the learned behaviour did not transfer to reliable generation. The explanation consistent with the evidence is exposure bias, not a lack of capacity. The methodological consequence is that speech-to-speech checkpoint selection should not rest on offline loss alone.

In the cascade track, two similar separations were seen. First, the 9/9 mechanical panel with the 0.5B responder does not establish the semantic quality of the 4B responder. Second, passing a same-family automatic test does not establish perceptual quality. The gap between a character error rate of 7.14% and a word error rate of 30.41% also shows that approximately preserving characters is not the same as word-level intelligibility.

The low score of the aligned reference in the semantic test is a reminder that the corpus's next turn is not necessarily an appropriate reply to the user's last utterance; this is why the LoRA adapter that reduced the development loss did not beat the baseline.

5.2 Why the Direct Track Failed

The final audit showed that Moshi's English tokenizer needs 6.6 tokens per Persian word and that 98.5% of turns exceed the one-token-per-frame budget. Removing this bottleneck with phoneme romanization raised the best generation result from 3/9 to 6/9 and made silent output audible, but the speech did not become intelligible; so the tokenizer bottleneck is real but not the only obstacle. J-Moshi [23] also replaced the tokenizer to bring Moshi to Japanese and then needed about 60,000 hours of dialogue pre-training, more than five hundred times this project's

data. The speech-input Gemma 4 model also showed that removing recognition, even with the same language model, lowers quality.

5.3 Limitations and Considerations

The latency and interruption evidence is automated and on an H100: the memory of a 12 GB card was respected but not its speed, and there was no human user, no physical microphone and loudspeaker path, and no real acoustic echo. The interruption and data labels are automatic and have no human labels; the two automatic judges of the responder disagree considerably; elderly speech in the tests is simulated. Interruption detection relies on the speech recognizer, not on the acoustic features specified in the project definition. The dataset is derived from an internal archive and is not redistributable. The results of the listening study have not yet been received.

5.4 Answers to the Research Questions

1. Low-rank adaptation of Moshika lowered held-out perplexity (text tokens 2.53 to 1.19, audio 13.98 to 4.71) but did not produce intelligible speech; the English tokenizer is the main obstacle. **2.** In the cascade, speculative end-of-speech detection with audio held until confirmation, streaming generation with a Persian-script constraint, clause chunking, a playback queue and two-stage interruption control with heard-only history brought the latency to a mean of 372 and a 90th percentile of 431 ms. **3.** For recognition and synthesis size did not buy quality: the smallest Persian-specialized recognizers and synthesizer gave the best trade-off, and the 7.7-billion-parameter end-to-end model did not reach intelligible speech. For the responder it did once the judges came from families with no candidate: Gemma 4 12B at four bits scored 1.2 to 1.75 points (of 10) above the best 4B with a 39 ms first token within 24 GB, but in the live stack its replies after late interruptions fell short of the coherence rule, so it is not qualified, and post-hoc 12 GiB checks of the same model failed it too. **4.** The perception of older adults depends on the results of the listening study. The status of each requirement is in Table 4.7.

5.5 Future Work

Completing the evidence: running the same loopback test on a 12–24 GB card with a physical audio path; collecting the listening-study responses and a live interaction session with older adults; human review of a sample of the labels; and human judging of the responders (Qwen3.5-4B with the elderly prompt has passed the deployment qualification, and the participants' preference decides on its replacement; whether people share the automatic judges' preference for the 24 GB responder has not been measured). Improving the system: a responder that knows the

recognizer can err (the 24 GB responder asked users to repeat after garbled follow-ups); learned separation of listener words from content words with a clipped onset (a 200 ms pre-roll was tested and rejected), semantic end-of-turn detection for the pauses of elderly speech (tested in two rounds; the best gate split fewer utterances but delayed too many finished turns, so a shorter hold or human-labelled turn ends are needed), and a more accurate recognizer within the same latency budget. Direct track: a tokenization that fits the frame budget, a remedy for the gap between teacher forcing and free generation, and the frozen-model approach.

Bibliography

[1] A. Défossez, L. Mazaré, M. Orsini, A. Royer, P. Pérez, H. Jégou, E. Grave, and N. Zeghidour. Moshi: A speech-text foundation model for real-time dialogue. Technical report, Kyutai, 2024. arXiv:2410.00037.

[2] X. Wang, Y. Li, C. Fu, Y. Zhang, Y. Shen, L. Xie, K. Li, X. Sun, and L. Ma. Freeze-Omni: A smart and low latency speech-to-speech dialogue model with frozen LLM. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 63345–63354. PMLR, 2025.

[3] M. F. Qharabagh, Z. Dehghanian, and H. R. Rabiee. ManaTTS Persian: A recipe for creating TTS datasets for lower-resource languages. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 9177–9206, Albuquerque, New Mexico, 2025. Association for Computational Linguistics.

[4] J. Xu, Z. Guo, J. He, H. Hu, T. He, S. Bai, K. Chen, J. Wang, Y. Fan, K. Dang, B. Zhang, X. Wang, Y. Chu, and J. Lin. Qwen2.5-Omni technical report. *arXiv preprint arXiv:2503.20215*, 2025.

[5] Z. Xie and C. Wu. Mini-Omni2: Towards open-source GPT-4o with vision, speech and duplex capabilities. *arXiv preprint arXiv:2410.11190*, 2024.

[6] Q. Fang, Y. Zhou, S. Guo, S. Zhang, and Y. Feng. LLaMA-Omni 2: LLM-based real-time spoken chatbot with autoregressive streaming speech synthesis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 18617–18629, Vienna, Austria, 2025. Association for Computational Linguistics.

[7] Kyutai Labs. Moshi-finetune: LoRA fine-tuning for Moshi. `https://github.com/kyutai-labs/moshi-finetune`, 2024.

[8] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. In *Proceedings of the 10th International Conference on Learning Representations*, 2022.

[9] M. Ranzato, S. Chopra, M. Auli, and W. Zaremba. Sequence level training with recurrent neural networks. In *Proceedings of the 4th International Conference on Learning Representations*, 2016.

[10] S. Bengio, O. Vinyals, N. Jaitly, and N. Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*, volume 28, pages 1171–1179, 2015.

[11] O. Kuchaiev, J. Li, H. Nguyen, O. Hrinchuk, R. Leary, B. Ginsburg, S. Kriman, S. Beliaev, V. Lavrukhin, J. Cook, and Others. NeMo: A toolkit for building AI applications using neural modules, 2019. arXiv:1909.09577.

[12] Rhasspy Project. Piper: A fast, local neural text-to-speech system. https://github.com/rhasspy/piper, 2023.

[13] S. B. Davis and P. Mermelstein. Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 28(4):357–366, 1980.

[14] A. de Cheveigné and H. Kawahara. YIN, a fundamental frequency estimator for speech and music. *The Journal of the Acoustical Society of America*, 111(4):1917–1930, 2002.

[15] J. H. Friedman. Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5):1189–1232, 2001.

[16] E. B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.

[17] G. Skantze. Turn-taking in conversational systems and human-robot interaction: A review. *Computer Speech & Language*, 67:101178, 2021.

[18] R. Roy, J. Raiman, S.-g. Lee, T.-D. Ene, R. Kirby, S. Kim, J. Kim, and B. Catanzaro. PersonaPlex: Voice and role control for full duplex conversational speech models, 2026. arXiv:2602.06053.

[19] International Telecommunication Union. Methods for subjective determination of transmission quality. *ITU-T Recommendation P.800*, 1996.

[20] A. Yang, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Li, D. Liu, F. Huang, H. Wei, H. Lin, J. Yang, J. Tu, J. Zhang, J. Yang, J. Zhou, J. Lin, K. Dang, and Others. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

[21] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv, and Others. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

[22] Daily. Smart Turn v3: Open-source semantic turn detection, 2025. Model `pipecat-ai/smart-turn-v3`, BSD-2-Clause.

[23] A. Ohashi, S. Iizuka, J. Jiang, and R. Higashinaka. Towards a Japanese full-duplex spoken dialogue system. In *Proceedings of Interspeech 2025*, 2025. arXiv:2506.02979.